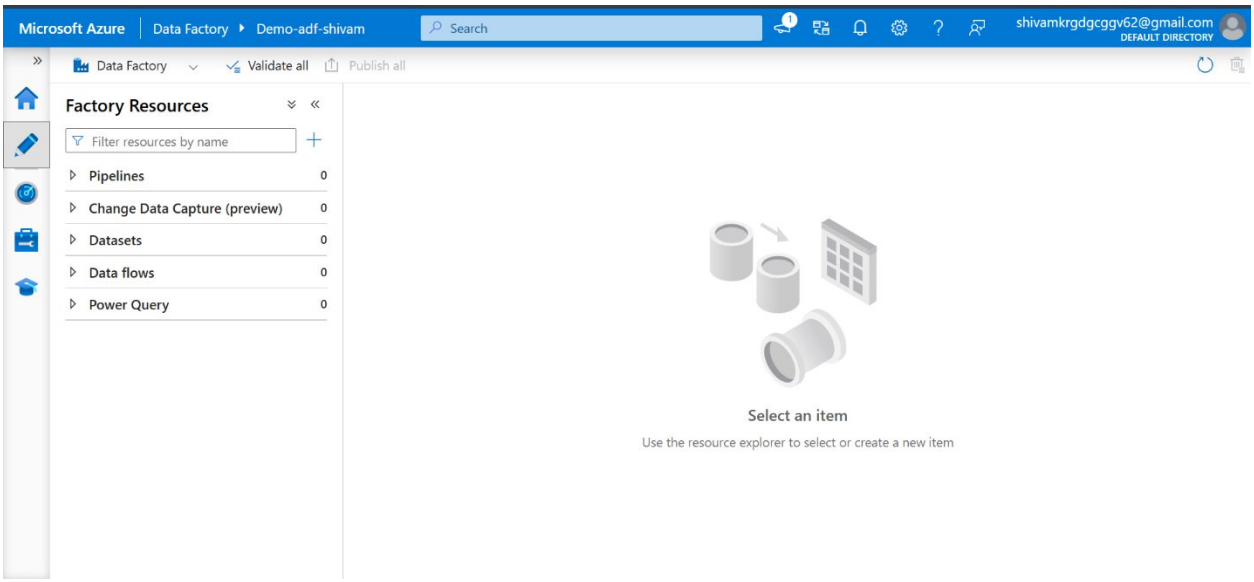


SESSION 5 — AZURE II: DATA MONITORING

- **Topic:** Azure II — Data Monitoring
- **Date:** 27th October 2026

14. Azure Monitor: Metrics, Logs, and Alerts

Azure Monitor is the service that watches your Azure resources. It collects signals from them, stores those signals, lets you look at them in charts and queries, and warns you when something goes wrong. For a data platform it answers questions such as "did last night's pipeline fail?" and "who read this secret?".

Signal	**Simple meaning**
Metrics	Numbers measured at regular intervals, such as used capacity of a storage account, failed pipeline runs, or CPU percentage. They are light, near real time, and collected automatically for most Azure resources. Platform metrics are kept for 93 days and are viewed in Metrics Explorer.
Activity log	The record of actions taken on a resource, such as who created, changed, or deleted it. It is collected automatically and kept for 90 days.
Resource logs	Detailed records of what happens inside a resource, such as each blob that was read or each pipeline run. They are not collected until you create a diagnostic setting.
Log Analytics workspace	The place where logs are stored and searched. It is queried with Kusto Query Language (KQL), and storage is charged by the amount of data ingested and kept.
Point	**Metrics vs. logs**

What it holds	Metrics: a number with a time stamp. Logs: a record with text and several fields.
Best for	Metrics: quick charts and fast alerts, such as failed pipeline runs above zero. Logs: finding the reason, such as which pipeline, which activity, and which error message.
Where to read	Metrics: Metrics Explorer. Logs: the Logs page of a Log Analytics workspace, using KQL.
Cost	Metrics: platform metrics are collected at no charge. Logs: you pay for the data sent to the workspace and kept there.

Azure Monitor: from signal to alert to action

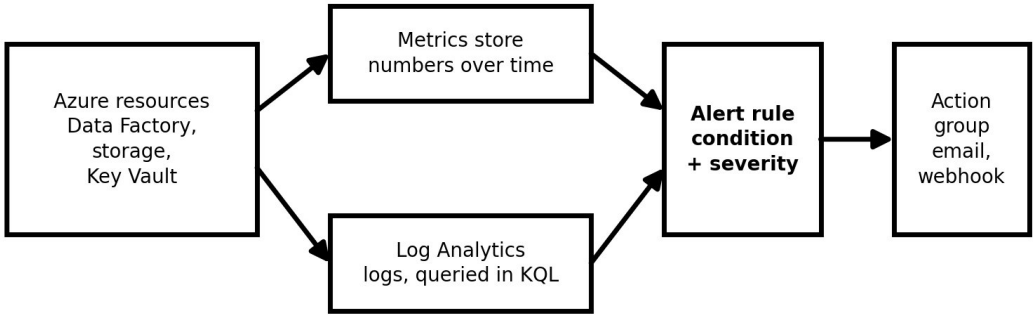


Diagram 11: Resources send metrics and logs to Azure Monitor, alert rules watch them, and action groups tell people.

Turning on logs. A resource sends its resource logs somewhere only after a diagnostic setting is created for it.

Step	**What to do**
1. Create a workspace	Create a Log Analytics workspace in the resource group, for example log-tricloud-dev.
2. Open the resource	Open the Data Factory, the Key Vault, or the storage account (its blob service) and go to Diagnostic settings, then Add diagnostic setting.
3. Pick categories	Choose only the log categories you need, such as PipelineRuns, ActivityRuns, and TriggerRuns for Data Factory, or AuditEvent for Key Vault.
4. Pick a destination	Send them to the Log Analytics workspace. A storage account is a cheaper archive, and Event Hubs streams the logs to other tools.

5. Query	After a few minutes, open the workspace and write a KQL query, as in the example below.

```
// Failed Data Factory pipeline runs in the last day
ADFPipelineRun
| where TimeGenerated > ago(1d)
| where Status == "Failed"
| project TimeGenerated, PipelineName, RunId, Status
| order by TimeGenerated desc
```

Alerts. An alert rule says: watch this, and when this condition is true, do that. It has four parts.

Part	**Simple meaning**
Scope	The resource being watched, such as the Data Factory or the storage account.
Condition	The signal and the test, such as failed pipeline runs greater than 0 within 5 minutes.
Severity	From Sev 0 (critical) to Sev 4 (verbose). It helps decide how fast someone must react.
Action group	A reusable list of who is told and what runs: email, SMS, push message, voice call, webhook, Azure Function, or Logic App.
Alert type	**Fires when**
Metric alert	A metric crosses a threshold, such as failed pipeline runs above 0 or storage availability below a set level.
Log search alert	A KQL query returns rows or a count above a limit, such as more than five failed requests in ten minutes.
Activity log alert	A management action happens, such as a resource being deleted, or Azure reports a service health event.

After an alert fires. It appears under Alerts in Azure Monitor with its severity. A metric alert is marked resolved when the condition clears, and a person can set the alert to acknowledged or closed to show who is handling it.

- **Alert on what needs a person.** Failed pipeline runs, a Databricks job that did not finish, or access denied on a Key Vault are good signals. Too many alerts teach people to ignore them all.
- **Audit the Key Vault.** Send the AuditEvent logs of the vault to the workspace, and a KQL query can show who read which secret and when (sections 10 and 11).
- **Reuse action groups.** Create one action group for the team and attach it to many rules. Alert processing rules can mute the alerts during planned maintenance.

- **Roles apply here too.** Monitoring Reader and Log Analytics Reader allow viewing, and Monitoring Contributor allows creating alerts. Give them at the right scope (section 7).
- **Mind the cost.** Logs are charged by the data ingested, so send only the categories that are needed and choose a sensible retention time.

Link to Databricks. A Data Factory alert tells you that the pipeline failed, and the Databricks job page shows why. Databricks workspaces on the Premium plan can also send their diagnostic logs to the same Log Analytics workspace, so that job and cluster activity can be searched with KQL.